WebClient Computing

Javascript Basics

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Announcements

- Homework-1 will be returned
- In Slack, send me a private message with **your KMU-ID and your nickname (no one can guess it is you!!!)** that will be used while I make public announcement about the homework/exam
- If you do not send me one, I will use your KMU-ID for announcements
- Homework-2 is out
  - This time, let's do it by Slack
    - Unexpected outcome of revealed homework before the due.
    - Project and further homework will be handled via GitHub
  - Do your homework, and **zip the files with KMU-ID: 20160000.zip**
  - Submit it via Slack private message

Exam Announcements

- Exam
  - April 22nd 10:30 ~ 11:30
  - Room: 7호관 231호
- Exam preparation
  - Focus on the lecture slide!
  - Do not memorize the tags/attributes but understand them
  - I will provide a list of tags/attributes

HTML/CSS Editor Recommendation

- http://brackets.io/
  - Handy for write HTML/CSS and check the output
- I need your inputs!!!
  - More than expected number of students did not hand in homework1
  - Is it too difficult?
  - Is the schedule too tight to complete?
  - Too busy to complete due to other course work?
  - Let me know using any method – slack, email, office-hour, …

Pseudo Element and Pseudo Class

- Pseudo element - effect to small portion (sub-part) of the element
  - ::first-line
    - Sets an effect to the first line of a block
  - ::first-letter
    - Sets an effect to the first character
  - ::before
    - Sets an effect before the tag happens
- Pseudo class – effect to the entire element
  - :last-child
    - Sets an effects to the last child element

Pseudo Example

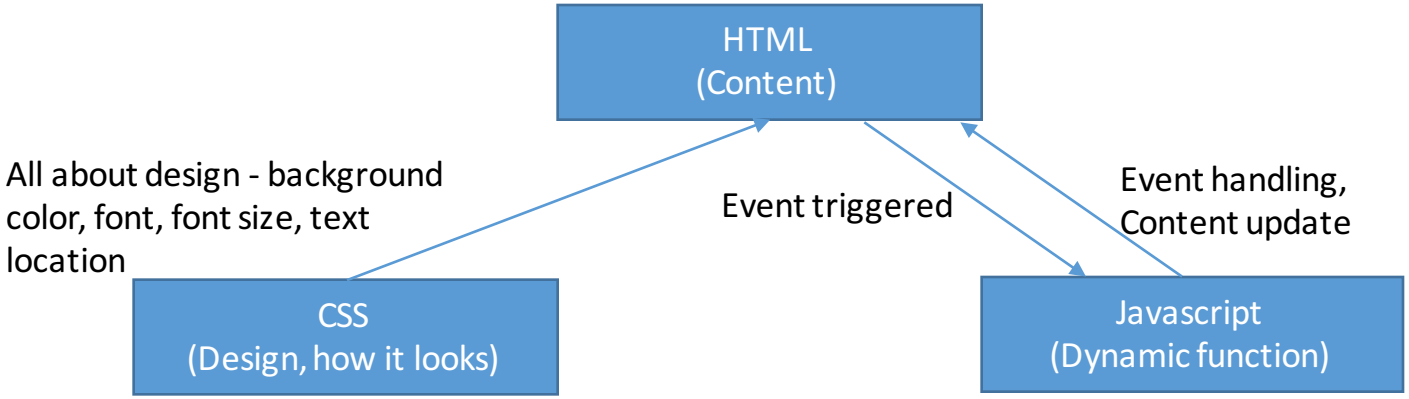
THIS IS THE FIRST SENTENSE. THIS IS THE NEXT SENTENCE. THEN A NEXT SENTENCE AND THEN A next sentence

This is a smiley heading

THIS IS THE LAST PARAGRAPH. IT WILL SHOW AN EFFECT OF P:LAST-CHILD WHERE THE EFFECT IS APPLIED ONLY TO THE LAST PARAGRAPH

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    p:last-child {
      color: green;
      text-transform: uppercase;
      font-style: italic;
    }
    p::first-letter {
      color: blue;
      font-size: xx-large;
    }
    p::first-line {
      color: red;
      font-variant: small-caps;
    }
    h1::before {
      content: url(smiley.gif);
    }
  </style>
</head>
<body>
  <p>This is the first sentense. This is the next sentence.
  Then a next sentence and then a next sentence</p>
  <h1>This is a smiley heading</h1>
  <p>This is the last paragraph. It will show an effect of
  p:last-child where the effect is applied only to the last paragraph</p>
</body>
</html>
```

HTML, CSS, and Javascript



- HTML – Hyper Text Markup Language
  - House itself
- CSS - Painting the house and decoration
- Javascript – rich features, electricity on/off using a smartphone

What is Javascript

- Javascript is an interpreted programming language with OO capabilities
- Most commonly used in web browsers
- Design goals
  - Enable interaction with users
  - Easily control web browser
  - Alter document in a web browser window
- Javascript language is well standardized and compatibility among browsers is good

Numbers in Javascript

- All numbers in 64 bit floating point
- Recognizes
  - Base-10 integer: 3421
  - Base-16 begins with 0X: 0XCAFE911
- Floating point literals
  - 3.14
  - .33333
  - 6.02e23 //$6.02*10^{23}$

Exercises

```
> var a=100;
<· undefined
> var b = 3.141592;
<· undefined
> var c = 3.2e2;
<· undefined
```

```
> typeof(a)
<· "number"
> typeof(b)
<· "number"
> typeof(c)
<· "number"
```

Type Conversion From Strings

- parseInt
- parseFloat

```
var num = prompt("Type a number to convert: ");
console.log(num, typeof(num));
var num_int = parseInt(num);
console.log(num_int, typeof(num_int));
var num_float = parseFloat(num);
console.log(num_float, typeof(num_float));
~
```

```
123.456 string                                  parse_int.js:2
123 "number"                                    parse_int.js:4
123.456 "number"                                parse_int.js:6
> |
```

String in Javascript

```
""  // The empty string: it has zero characters
'testing'
"3.14"
'name="myform"'
"Wouldn't you prefer O'Reilly's book?"
"This string\nhas two lines"
"π is the ratio of a circle's circumference to its diameter"
```

- Double quote string can be contained within a single quote string and vice versa

  > console.log(‘name=“myform”’)

  > consoel.log(“name=‘myform’”)

String Exercise

- string.js

```
var a = "String with double quote";
var b = 'string with single quote';
var c = 'double "quote" within single quote';
var d = "single 'quote' within double quote";
```

```
> console.log(a)
  String with double quote                                  VM288:2
<· undefined
> console.log(b)
  string with single quote                                  VM361:2
<· undefined
> console.log(c)
  double "quote" within single quote                        VM434:2
<· undefined
> console.log(d)
  single 'quote' within double quote                        VM487:2
<· undefined
>
```

Escape Character and String Concatenation

- Escape character
  - A predefined set of characters that comes after back slash for a specific purpose
    - \n new line
    - \t tab
    - Escaping \ - \\

Example

- escape.js

```
var escape_string = "this is\na new line\tthen have a tab\nnew line and \\escape";
console.log(escape_string);
```

```
this is                                                    escape.js:2
a new line  then have a tab
new line and \escape
```

Undefined, Null Variable, NaN

- Undefined variable
  - A variable name is defined but the value is not initialized yet
  - In an object, accessing an element that is not a member of an object
- Null variable
  - An empty object with a value null
  - User sets a variable to null to make the value as **empty**
- NaN
  - Acronym of Not a Number
  - parseInt(“string”)

Example

- undefine_null.js

```
var undef;

console.log(undef);
console.log(typeof(undef));

undef = null;
console.log(undef);
console.log(typeof(undef));

undef = parseInt("string");
console.log(undef);
```

```
undefined                    undefined_null.js:3
undefined                    undefined_null.js:4
null                         undefined_null.js:7
object                       undefined_null.js:8
NaN                          undefined_null.js:11
```

Reference

- Javascript, the definitive guide
- W3Schools Javascript